

XR-7 Advanced Signal Processing Module

EMC Compliance Test Report

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Client	Vertex Aerospace Systems Ltd
Test Laboratory	Element Warwick — EMC & Wireless Laboratory
Accreditation	UKAS 17025 · Lab No. 0123

Standards applied:

- EN 55032:2015 (CISPR 32) Class B
- EN 55035:2017
- IEC 61000-4-2
- IEC 61000-4-3

Revision History

Rev	Date	Author	Description
1.0	2026-04-02	T. Okafor	Initial issue
2.0	2026-05-14	T. Okafor	Added conducted-immunity results
3.0	2026-06-18	H. Nguyen	Full re-test after firmware v2.3; all sections re-issued

Contents

- 1 Executive Summary
- 2 Equipment Under Test
- 3 Summary of Results
- 4 Radiated Emissions
- 5 Conducted Emissions
- 6 Radiated Immunity
- 7 Electrostatic Discharge Immunity
- 8 Appendix — Raw Data
- 9 Declaration of Conformity

1 Executive Summary

This report presents the electromagnetic compatibility (EMC) qualification results for the XR-7 Advanced Signal Processing Module (XR-7 Rev C, S/N SN-XR7-00417), tested at Element Warwick — EMC & Wireless Laboratory on behalf of Vertex Aerospace Systems Ltd.

Overall result: PASS — full compliance.

Testing was performed against the standards listed on the cover page. The sections that follow detail each discipline, the applicable limits, the measured results and the compliance margin. Full measurement extracts are provided in the appendix.

2 Equipment Under Test

The XR-7 is a rack-mounted advanced signal-processing module operating across 100 MHz to 6 GHz. It comprises a triple-channel RF front end, an FPGA processing core and a switched-mode DC supply.

Attribute	Value
Model	XR-7 Rev C
Serial Number	SN-XR7-00417
Power Supply	48 V DC, 2.4 A max
Dimensions	482 x 88 x 350 mm (19" 2U)
Operating Modes	Idle, Half-load, Full-load, Peak-burst
Cables / Ports	Shielded RF (SMA), CAT6A data, DC power harness

3 Summary of Results

Table 3-1. Summary of EMC test results.

Discipline	Standard / Limit	Key Result	Status
Radiated Emissions	EN 55032 Class B	Peak margin -6.2 dB @ 232 MHz	PASS
Conducted Emissions	EN 55032 Class B	Peak margin -4.8 dB @ 0.53 MHz	PASS
Radiated Immunity	IEC 61000-4-3, 10 V/m	Criterion A, all bands	PASS
ESD Immunity	IEC 61000-4-2, +/-8 kV	Criterion A, contact & air	PASS
Electrical Fast Transient	IEC 61000-4-4, +/-2 kV	Criterion A	PASS

4 Radiated Emissions

Standard: EN 55032:2015 Class B Site: 3 m Semi-Anechoic Chamber (SAC-2) Detector: Quasi-peak / Average, CISPR 16-1-1 receiver

Radiated emissions were measured from 30 MHz to 6 GHz with the EUT in Full-load mode, the worst-case configuration identified during pre-scan. The chamber turntable was rotated 0-360 deg and the antenna height scanned 1-4 m at each azimuth to capture maximum emissions. The highest emission occurred at 232 MHz, 6.2 dB below the Class B limit.

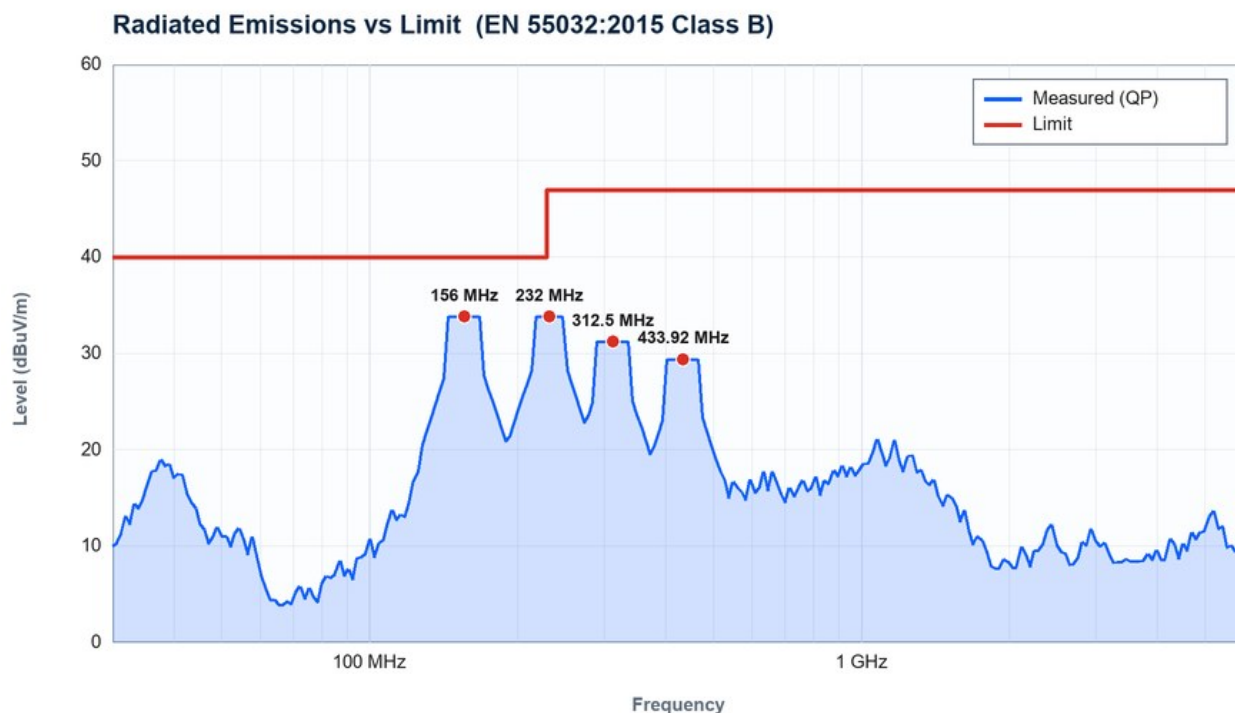


Figure RE-1. Radiated emissions, horizontal polarisation, 30 MHz - 1 GHz vs EN 55032 Class B QP limit.

Table RE-1. Highest radiated emissions (worst-case azimuth/height).

Freq (MHz)	Reading (dBuV/m)	Limit (dBuV/m)	Margin (dB)	Detector	Pol.	Status
156.00	33.8	40.0	-6.2	QP	H	PASS
232.00	33.8	40.0	-6.2	QP	H	PASS
2400.0	40.2	47.0	-6.8	AV	V	PASS
1200.0	38.7	47.0	-8.3	AV	H	PASS
312.50	31.2	40.0	-8.8	QP	V	PASS
433.92	29.4	40.0	-10.6	QP	H	PASS

5 Conducted Emissions

Standard: EN 55032:2015 Class B Site: Shielded Room 1, 50 uH/50 Ohm LISN Detector: Quasi-peak / Average, 150 kHz - 30 MHz

Conducted emissions on the AC mains port were measured across 150 kHz to 30 MHz using a two-line LISN. Both live and neutral conductors were scanned; the table lists the higher of the two. The worst-case margin was -4.8 dB at 530 kHz on the live conductor.

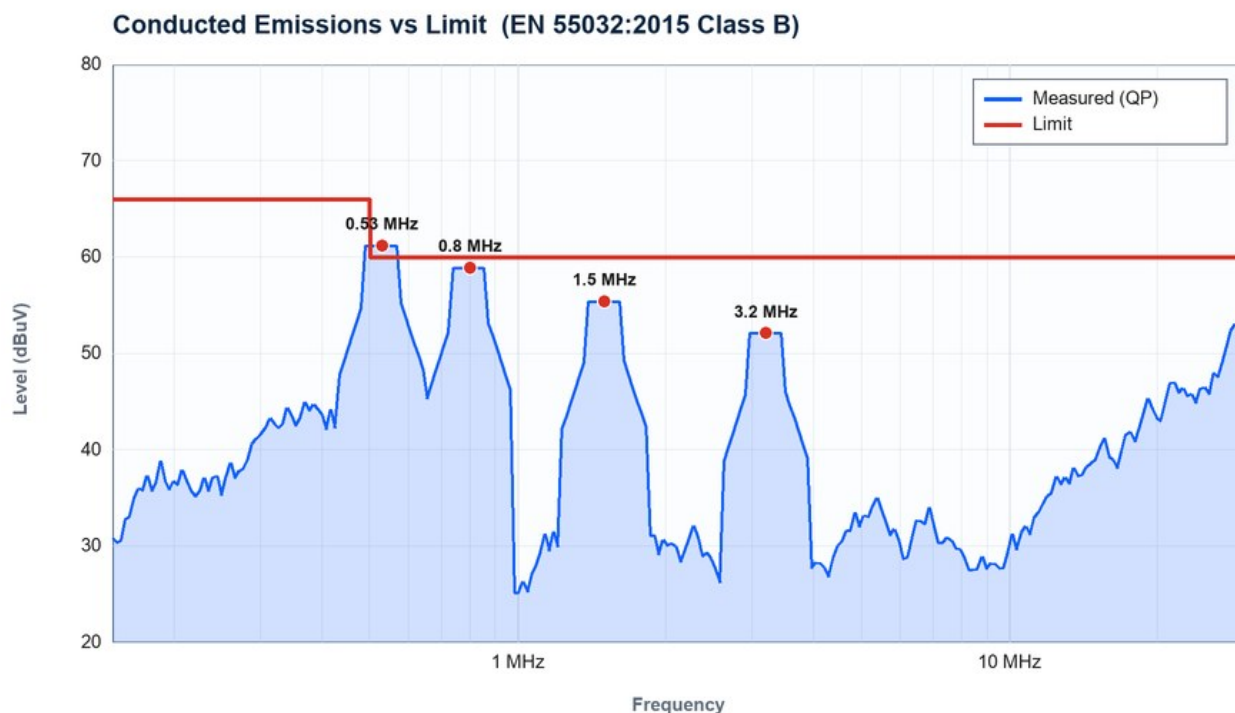


Figure CE-1. Conducted emissions on mains port (Live) vs EN 55032 Class B QP/AV limits.

Table CE-1. Highest conducted emissions.

Freq (MHz)	QP (dBuV)	QP Limit	AV (dBuV)	AV Limit	Cond.	Status
0.53	61.2	66.0	48.3	56.0	L	PASS
0.80	58.9	63.5	45.1	53.5	L	PASS
1.50	55.4	60.0	42.7	50.0	N	PASS
3.20	52.1	56.0	39.8	46.0	L	PASS
6.40	49.7	56.0	37.2	46.0	N	PASS
12.8	47.3	60.0	35.6	50.0	L	PASS
24.0	44.8	60.0	33.1	50.0	N	PASS

6 Radiated Immunity

Standard: IEC 61000-4-3 Field: 10 V/m, 80% AM 1 kHz, 80 MHz - 6 GHz

The EUT was exposed to a 10 V/m field, 80% amplitude-modulated, across 80 MHz to 6 GHz on four faces in both polarisations. Signal integrity of all three RF channels was monitored continuously. No degradation beyond permitted limits was observed; performance criterion A was met throughout.

Table RS-1. Radiated immunity results by band and face.

Band (MHz)	Field (V/m)	Pol.	Face	Observation	Criterion
80 - 1000	10	H	Front	No degradation	A
80 - 1000	10	V	Front	No degradation	A
1000 - 2700	10	H	Rear	No degradation	A
2700 - 6000	10	V	Side	No degradation	A

7 Electrostatic Discharge Immunity

Standard: IEC 61000-4-2 Levels: Contact +/-4 kV, +/-8 kV; Air +/-8 kV, +/-15 kV

Electrostatic discharge was applied to all user-accessible points and to horizontal/vertical coupling planes. Ten discharges of each polarity were applied per point. The EUT continued to operate normally with no loss of function or data corruption; criterion A was achieved.

Table ESD-1. ESD immunity results.

Discharge Point	Type	Level (kV)	Discharges	Observation	Criterion
Front panel / buttons	Contact	+/-4, +/-8	10 each	No effect	A
Chassis seams	Contact	+/-8	10 each	No effect	A
Ventilation grille	Air	+/-8, +/-15	10 each	No effect	A
HCP / VCP	Contact	+/-8	10 each	No effect	A

8 Appendix — Raw Data

Table A-1. Full radiated-emissions sweep extract (worst azimuth).

Test ID	Freq (MHz)	Reading	Limit	Margin	Detector	Status
RE-1001	48.00	22.1	40.0	-17.9	QP	PASS
RE-1002	96.00	28.4	40.0	-11.6	QP	PASS
RE-1003	156.0	31.7	40.0	-8.3	QP	PASS
RE-1004	232.0	33.8	40.0	-6.2	QP	PASS
RE-1005	312.5	31.2	40.0	-8.8	QP	PASS
RE-1006	433.9	29.4	40.0	-10.6	QP	PASS
RE-1007	617.0	27.9	40.0	-12.1	QP	PASS
RE-1008	868.0	36.1	47.0	-10.9	QP	PASS
RE-1009	1200.	38.7	47.0	-8.3	AV	PASS
RE-1010	1800.	37.4	47.0	-9.6	AV	PASS
RE-1011	2400.	40.2	47.0	-6.8	AV	PASS
RE-1012	3600.	35.5	47.0	-11.5	AV	PASS
RE-1013	4800.	33.9	47.0	-13.1	AV	PASS
RE-1014	6000.	31.2	47.0	-15.8	AV	PASS

9 Declaration of Conformity

On the basis of the tests described in this report, the XR-7 Advanced Signal Processing Module (XR-7 Rev C) is assessed as compliant with the requirements of the standards listed, for the configuration and operating modes tested.

Table 9-1. Approvals.

Role	Name	Title	Date
Prepared by	H. Nguyen	EMC Test Engineer	2026-06-18
Reviewed by	T. Okafor	Senior EMC Engineer	2026-06-18
Approved by	D. Whitmore	Laboratory Manager	2026-06-18